

Temporal Measurement Instability in LLM Benchmarks: A Large-Scale Psychometric Audit of 5,227 Models

Anonymous ACL submission

Abstract

LLM benchmark scores are routinely compared across model generations, yet whether individual items measure the same construct over time is rarely tested—and existing contamination detectors validate against ground truth that conflates global exposure with differential functioning. We develop a semi-synthetic injection framework with cross-subject matching that achieves zero false positive inflation ($\Delta\text{FPR} = 0.000$) while preserving detection power ($\Delta\text{TPR} = +0.508$ at $p = 0.5$), and show that web-overlap contamination labels are near-independent of Mantel-Haenszel Differential Item Functioning (DIF) flags (enrichment ≈ 0.96 , $\phi = 0.012$)—exposing a category error in prevailing validation methodology. Applying MH-DIF with ETS severity classification to six METABENCH benchmarks (5,227 models, 27,125 items), we find 17.9–31.3% of items exhibit large temporal DIF (ETS C) between 2023 and 2024 cohorts, each 45–174 $\times$ above random baselines, with knowledge/reasoning benchmarks more affected ($C\% \geq 26.9\%$) than procedural ones ($C\% \leq 18.9\%$). This does not threaten ranking validity ($\rho \approx 0.997$) but reveals which capability dimensions shift across generations. Item-level decomposition shows 92.4% of the signal is item-specific; DIF-flagged items concentrate discriminative information ($z = -8\sigma$ to -49σ) with a difficulty–direction interaction ($\rho = -0.597$): gains on hard items, erosion on easy ones. LLM benchmarks need psychometric monitoring, not just score reporting.

1 Introduction

When we say “Model B outperforms Model A on MMLU,” we implicitly assume that the benchmark measures the same construct in the same way for both models. This assumption of temporal comparability underpins every cross-generation comparison published on leaderboards and in model reports, yet it is rarely tested. We apply Mantel-

Haenszel Differential Item Functioning (MH-DIF) analysis (Holland and Thayer, 1988) with Educational Testing Service (ETS) severity classification to the METABENCH response matrix (Kipnis et al., 2024),¹ and find that 31% of MMLU (Hendrycks et al., 2021) items receive ETS Category C classification ($|\Delta_{\text{MH}}| \geq 1.5$, $p < 0.05$), indicating large shifts in item functioning between models released in 2023 and 2024. This is not statistical noise: a random-split control within the 2023 cohort yields only 0.18% ETS C items. This pattern persists across all 57 MMLU subjects ($C\%$ range 21–48%, unweighted mean $\approx 30\%$), and item-level regression shows that 92.4% of the DIF signal is unrelated to aggregate domain-level accuracy gains. Nearly one in three MMLU items behaves differently depending on when the evaluated models were trained. Crucially, this measurement instability does not threaten benchmark validity: Spearman rank correlation between model rankings computed on all items versus stable-only items is $\rho \approx 0.997$. The instability is diagnostic, not destructive; it reveals which capability dimensions shift across model generations rather than rendering MMLU scores unreliable.

A growing body of contamination detection research has produced sophisticated item-level methods, from membership inference approaches such as Min-K% (Shi et al., 2024) to performance-based detectors such as ConStat (Dekoninck et al., 2024), DVD (Liang, 2026), and Self-Critique (Tao et al., 2026). These methods validate detection accuracy primarily against web-overlap labels such as those of Li et al. (2024b), which classify benchmark items as contaminated based on verbatim or near-verbatim matches in publicly accessible training corpora. Such labels measure *global exposure*: whether an item exists in the data ecosys-

¹Analysis code and DIF audit scripts are available at <https://anonymous.4open.science/r/dif-audit>.

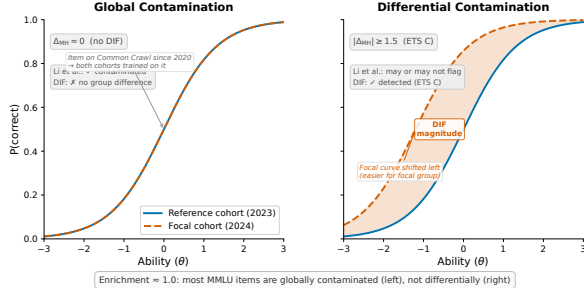


Figure 1: Global vs. differential contamination, illustrated via Item Characteristic Curves (ICCs). **Left:** global contamination produces no DIF; both cohorts encounter the item, and their ICCs overlap. **Right:** differential contamination produces DIF; the focal cohort’s ICC shifts because only that cohort was disproportionately exposed during training. The enrichment ratio of ≈ 1.0 between DIF flags and web-overlap labels is expected because most MMLU items fall into the left category.

tem that all training pipelines can access. They do not measure *differential item functioning*: whether an item behaves differently across model cohorts that may have had varying degrees of exposure or different training methodologies. Figure 1 illustrates the distinction. Most MMLU items are globally contaminated in the sense that they exist in web-accessible text, so all model cohorts encounter them and no differential signal arises. The items that function differentially are those where exposure or training methodology changed asymmetrically across cohorts. Validating differential detectors against global-exposure labels therefore conflates two phenomena. Direct comparison confirms near-independence between the two flag sets (enrichment = 0.964, Jaccard = 0.206, $\phi = 0.012$), consistent across contamination subtypes.

We apply classical MH-DIF with ETS classification to the full METABENCH response matrix (5,227 models $\times$ 12,508 MMLU items), to our knowledge, the largest-scale DIF analysis on LLM benchmarks. This psychometric approach complements concurrent work on anchor-item calibration for score comparability (Habba et al., 2026) and addresses the fragility of existing contamination detectors under post-training perturbation (Wang et al., 2026). Semi-synthetic injection experiments confirm that MH-DIF is sensitive to differential item behavior when present ($\Delta\text{TPR} = +0.508$ at exploitation rate $p = 0.5$), and our cross-subject matching protocol eliminates false positive inflation ($\Delta\text{FPR} = 0.000$). Three structural

controls (matched-ability quintile analysis, IRT θ -conditioning, and within-subject pre-checks) confirm that the approximately 31% temporal DIF rate cannot be attributed to methodological artifacts, establishing it as a robust empirical finding about LLM benchmark measurement properties, replicated across all six METABENCH benchmarks (Section 4.7). MH-DIF detects *that* items function differently across cohorts, not *why*—the signal may reflect contamination, capability change, or training methodology shifts.

MMLU’s 12,508 items provide sufficient measurement redundancy that removing nearly a third does not alter rank order ($\rho \approx 0.997$). DIF should therefore be understood as a diagnostic lens rather than a quality filter: DIF-flagged items concentrate discriminative signal ($z = -8\sigma$ to -49σ vs. random removal), and model families show distinctive DIF profiles (Llama-3 gains +53 median rank positions, Mistral loses 35).

Our contributions are methodological and empirical, not algorithmic:

- Semi-synthetic validation framework:** Controlled injection with cross-subject matching achieves $\Delta\text{FPR} = 0.000$ while preserving detection power ($\Delta\text{TPR} = +0.508$ at $p = 0.5$); cross-benchmark replication on GSM8K confirms generality (Sections 4.5 and 4.7).
- Ground truth meta-critique:** Web-overlap contamination labels (Li et al., 2024b) do not provide appropriate ground truth for item-level differential detectors. Direct empirical comparison on 9,229 items shows near-independence (enrichment = 0.964, Jaccard = 0.206, $\phi = 0.012$), exposing a validation methodology gap (Section 4.3).
- Temporal measurement instability at scale:** Approximately 31% of MMLU items exhibit differential functioning across model generations under ETS C classification; replication across all six METABENCH benchmarks yields $C\% = 17.9\text{--}31.3\%$ ($45\text{--}174\times$ above random baselines), with knowledge/reasoning benchmarks showing higher DIF rates than procedural ones. DIF-flagged items concentrate discriminative signal ($z = -8\sigma$ to -49σ vs. random removal) and exhibit a difficulty-direction interaction ($\rho = -0.597$): gains on hard items, erosion on easy ones (Sections 4.2, 4.6 and 4.7).

2 Related Work

Contamination Detection. A growing family of methods detects benchmark contamination through model behavior or training data analysis. Membership inference approaches estimate whether a model has seen specific items during training: Min-K% (Shi et al., 2024) and Min-K%++ (Zhang et al., 2024) use token-level log-likelihood statistics, while Time Travel (Golchin and Surdeanu, 2024) probes models with temporal cues. Performance-based detectors identify contamination through statistical anomalies: ConStat (Dekoning et al., 2024) tests performance gaps between original and perturbed items, DVD (Liang, 2026) uses distributional verification, and Self-Critique (Tao et al., 2026) employs model self-evaluation. Balloccu et al. (2024) provide a broader taxonomy of leakage detection strategies. These methods share a common validation practice: they verify detection accuracy against web-overlap or n-gram labels (Li et al., 2024b) that classify items based on verbatim matches in publicly accessible corpora. Such labels measure *global exposure* (whether an item exists in training data) rather than *differential item functioning* (whether an item behaves differently across model cohorts). This global-vs-differential distinction motivates our ground truth analysis (Section 4.3). Work on LLM memorization (Carlini et al., 2023; Tirumala et al., 2022) establishes that memorization depends on data frequency and model scale, reinforcing the distinction between global exposure and differential behavior. Wang et al. (2026) show that existing contamination detectors are fragile under reinforcement learning, further motivating behavioral approaches grounded in psychometric theory.

IRT in LLM Evaluation. IRT adoption in LLM evaluation has accelerated: Lalor et al. (2019) introduced IRT to NLP, Vania et al. (2021) assessed test set discriminability, and Rodriguez et al. (2021) demonstrated that not all evaluation examples are equally informative. Recent extensions include PSN-IRT (Zhou et al., 2026) for benchmark quality at scale, TinyBenchmarks (Maia Polo et al., 2024) for IRT-based item selection, scalability coefficients for detecting bad benchmark items (Hardy et al., 2026), ATLAS (Li et al., 2025) for computerized adaptive testing of LLMs, and competency-focused IRT evaluation in medical domains (Luo et al., 2025). The closest concurrent work is Growing Pains (Habba et al., 2026), which addresses

score comparability through fixed-parameter calibration (FPC; Kim and Kolen, 2019). Our work is complementary: Growing Pains assumes anchor item stability, while DIF analysis diagnoses which items’ measurement properties shift across model generations, potentially flagging anchor degradation before it affects calibration accuracy. Empirical comparison awaits public release of anchor item lists.

DIF Methodology and Measurement Invariance. The Mantel-Haenszel procedure for DIF detection was introduced by Holland and Thayer (1988) and remains the most widely used method in educational testing due to its simplicity and well-characterized statistical properties (Clauser and Mazor, 1998). Alternative approaches include logistic regression DIF (Swaminathan and Rogers, 1990), SIBTEST (Shealy and Stout, 1993), Lord’s χ^2 test (Lord, 1980), and the Raju area method (Raju, 1988); Zumbo (1999) provides a comprehensive comparison. DIF analysis is a specific instance of measurement invariance testing (Meredith, 1993; Vandenberg and Lance, 2000), which examines whether a measurement instrument functions equivalently across populations. MIMIC models offer a latent-variable approach to the same problem (Muthén, 1985; Woods, 2009), and Teresi and Fleishman (2007) reviews DIF methods across applied domains. Cross-subject matching—computing the stratification variable from domains other than the one being tested—has psychometric precedent in the DIF literature (Nandakumar, 1993; Donoghue and Allen, 1993); our adaptation addresses LLM-specific challenges of multi-domain local dependence and simultaneous contamination control at scale. We do not modify the MH procedure; our contribution is the application context and the empirical findings it reveals. The scale of this analysis (5,227 examinees across a 57-subject multidimensional ability space) exceeds typical educational testing applications, where Belzak (2020) note that DIF methods can behave differently at extreme sample sizes. The primary methodological challenge is the impact-versus-DIF confound: genuine ability differences between temporal cohorts can inflate DIF estimates. We address this through matched-ability quintile analysis, IRT θ -conditioning, and a random-split control (Section 4.2).

Benchmark Quality and Design. The Benchmark Health Index (Zhu et al., 2026) and

Benchmark² (Qian and Huang, 2026) assess static benchmark quality; our temporal DIF analysis is complementary, diagnosing *when* measurement properties shift rather than evaluating them at a single snapshot. Maeda (2025) predict which item features associate with DIF in educational settings; we instead detect DIF empirically across LLM cohorts at scale. Dynamic benchmarks (Kiela et al., 2021; Li et al., 2024a; White, 2025) mitigate contamination by construction, yet none provides psychometric equating across versions. Chatbot Arena (Chiang et al., 2024) sidesteps static benchmarks through pairwise preference. Broader surveys (Chang et al., 2024; Burnell et al., 2023) identify benchmark saturation and lack of item-level transparency as systemic challenges that our DIF analysis directly addresses.

3 Method

We adopt Differential Item Functioning (DIF) analysis from educational measurement to assess whether individual benchmark items behave consistently across groups of LLMs. The procedure requires only a binary response matrix and group labels, with no access to model internals or training data. Below we define the data, the statistical test, the cohort designs, and a semi-synthetic validation framework.

3.1 Data

The primary dataset is the METABENCH response matrix (Kipnis et al., 2024), which records binary outcomes (1 = correct, 0 = incorrect) for 5,227 models evaluated on 12,508 MMLU items (Hendrycks et al., 2021). Models span release dates from 2023 to 2024, drawn from the Open LLM Leaderboard. MMLU covers 57 subjects grouped into five broad domains (STEM, Humanities, Social Sciences, Other, and a mixed category). We supplement the response matrix with the web-overlap contamination labels of Li et al. (2024b), which classify 9,229 of 12,508 items (73.8% coverage) as “contaminated” or “clean” based on verbatim matches in publicly accessible web corpora. These labels serve as an external reference for evaluating the relationship between DIF flags and existing contamination indicators.

3.2 Mantel-Haenszel DIF and ETS Classification

In educational testing, Differential Item Functioning (DIF) occurs when an item’s statistical relation-

ship to the latent trait θ differs between two groups after conditioning on ability level. An item with DIF is not merely harder for one group; it functions differently relative to what the group’s overall ability would predict. The standard terminology refers to test-takers as “examinees” and questions as “items.” In our setting, examinees are LLMs and items are benchmark questions.

The Mantel-Haenszel (MH) procedure (Holland and Thayer, 1988) detects DIF by stratifying examinees into K score bins based on total score (a proxy for θ) and computing a common odds ratio across strata. We use $K = 5$ equiprobable strata (quintile-based binning) following ETS recommended practice for large-scale assessments (Donoghue and Allen, 1993; Clauser and Mazor, 1998). Within each stratum k , the responses of the reference group and the focal group form a 2×2 contingency table:

$$\alpha_{\text{MH}} = \frac{\sum_{k=1}^K A_k D_k / T_k}{\sum_{k=1}^K B_k C_k / T_k}, \quad (1)$$

where A_k and B_k are the number of correct and incorrect responses in the reference group for stratum k , C_k and D_k are the corresponding counts for the focal group, and T_k is the total number of examinees in stratum k . Under the null hypothesis of no DIF, $\alpha_{\text{MH}} = 1$. The log odds ratio is converted to the ETS delta scale:

$$\Delta_{\text{MH}} = -2.35 \ln(\alpha_{\text{MH}}). \quad (2)$$

The Educational Testing Service (ETS) classifies DIF severity into three categories based on $|\Delta_{\text{MH}}|$. Category A (negligible) requires $|\Delta_{\text{MH}}| < 1.0$. Category C (large) requires both $|\Delta_{\text{MH}}| \geq 1.5$ and statistical significance at $p < 0.05$ from the MH chi-squared test. Category B (moderate) includes all remaining items: those with $1.0 \leq |\Delta_{\text{MH}}| < 1.5$, or $|\Delta_{\text{MH}}| \geq 1.5$ without statistical significance. The dual threshold in ETS C (effect size *and* significance) guards against flagging items that show large but unreliable effect sizes. Throughout this paper, we report the percentage of items classified as ETS C, denoted C%. For example, if 2024 models consistently outperform 2023 models on a formal-logic item *after controlling for overall ability*, the pooled odds ratio $\alpha_{\text{MH}} = 0.42$

yields $\Delta_{\text{MH}} = -2.35 \ln(0.42) = 2.04$, classified as ETS C ($|\Delta_{\text{MH}}| \geq 1.5, p < 0.001$).

The choice of matching variable (the total score used for stratification) affects both validity and interpretation. We distinguish three matching strategies: (1) *standard matching* uses the within-benchmark total score on all items of the same benchmark; (2) *cross-subject matching* uses the total score on all subjects *other than* the one containing the item under test, preventing the matching variable from absorbing injected contamination signals (Section 3.4); and (3) *external matching* uses the total score on a different benchmark entirely (e.g., MMLU total score as the matching variable when testing DIF on GSM8K items), providing an independent ability proxy uncontaminated by the target benchmark. Cross-subject and external matching both address matching variable contamination but at different granularities: cross-subject operates within a multi-domain benchmark, while external operates across benchmarks. Total score matching is sufficient: IRT θ estimates correlate at $r = 0.9986$ with total scores in our data, confirming that computationally simpler CTT-based stratification introduces no bias relative to IRT ability matching.

LLM-specific challenges. Applying MH-DIF to LLMs introduces assumptions absent in educational testing. First, models trained on overlapping corpora violate local independence; permutation simulation (1,000 iterations $\times$ 8 subjects, Q3 = 4.6–34.2%) confirms LD-attributable FPR inflation < 0.001 pp under cross-subject matching with ETS C (Section A). Second, fine-tuned variants and merges within model families are not independent examinees; graduated de-duplication shows family structure does not inflate C% (Section 4.4). Third, the ability distribution is non-normal (bimodal from base vs. instruction-tuned models), motivating quintile-based rather than parametric stratification. Fourth, MMLU’s 57 subjects span multiple latent dimensions, while MH assumes unidimensionality; cross-subject matching and within-subject DIF checks address this (Section 4.4). MH-DIF is non-parametric and does not assume a specific IRT model.

3.3 Cohort Designs

We construct five cohort comparisons: (1) *Temporal*: 2023 ($N = 1,080$) vs. 2024 ($N = 807$) models, the primary comparison; (2) *Ability-quintile*:

Q1–Q2 vs. Q4–Q5 within the same temporal window, testing whether ability differences alone produce DIF; (3) *Within-family*: Llama-2 ($N = 161$) vs. Llama-3 ($N = 308$), isolating generational changes within a single family; (4) *Within-subject*: per-subject DIF to test domain concentration; and (5) *Random split*: two random halves of the 2023 cohort ($N = 540$ each) as a sanity check.

3.4 Semi-Synthetic Injection Framework

To evaluate whether MH-DIF can detect differential contamination when it is present, we construct a semi-synthetic framework with controlled ground truth.

Procedure. From items classified as ETS A in the uninjected data (negligible DIF), we select 20% uniformly at random as the “contaminated” set. For each contaminated item, we flip incorrect responses to correct in the focal group at exploitation rate $p \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$. We then run MH-DIF on the modified response matrix and measure detection performance. We report $\Delta\text{TPR} = \text{TPR}(\text{injected}) - \text{TPR}(\text{baseline})$ and $\Delta\text{FPR} = \text{FPR}(\text{injected}) - \text{FPR}(\text{baseline})$, where TPR is the true positive rate on injected items and FPR is the false positive rate on uninjected items. Each injection level is repeated across 5 random seeds for uncertainty estimation.

Cross-subject matching. The choice of matching variable is critical for valid semi-synthetic evaluation. When contamination is injected into items within subject X , the total score on subject X absorbs part of the injection signal. Using this contaminated total score as the MH matching variable violates the conditional independence assumption and inflates FPR. Cross-subject matching resolves this by computing the matching variable from all subjects *other than* the one being tested.

Remark (Zero FPR under single-subject matching). Let $M_i^{(-s)} = \sum_{j \neq s} Y_{ij}$ be the matching variable for examinee i on focal subject s . Under single-subject injection (contamination confined to subject s), the tested item belongs to s and does not contribute to $M_i^{(-s)}$; injecting contamination into s leaves $M_i^{(-s)}$ unchanged. Therefore $\Delta\text{FPR} = 0$ by construction.

When contamination spans multiple subjects simultaneously, $M_i^{(-s)}$ may absorb signal from other contaminated subjects, potentially inflating

FPR. Empirically, this guarantee extends to multi-domain contamination affecting $\leq 9\%$ of subjects ($k \leq 5$), with graceful degradation beyond this boundary (Section 4.5).

Empirically, $\Delta\text{FPR} = 0.000$ across all five MMLU domains (bootstrap 95% CI = $[0.000, 0.000]$, 5,000 resamples), while detection power is preserved ($\Delta\text{TPR} > 0.3$ at $p = 0.5$ for four of five subjects). We adopt cross-subject matching as the standard protocol for all semi-synthetic analyses.

Diagnostic procedures (Yen’s Q3 local independence, domain-specificity analysis, DIF decomposition) and statistical details (multiple testing correction, threshold sensitivity, architecture heterogeneity checks) appear in Section A.

4 Experiments

We apply MH-DIF to the METABENCH response matrix (Kipnis et al., 2024) (5,227 models $\times$ 12,508 MMLU items (Hendrycks et al., 2021)), with temporal cohorts defined by release date (2023 vs. 2024). Cross-benchmark validation across all six METABENCH benchmarks appears in Section 4.7.

4.1 Validation: DIF Detection Power

A Monte Carlo power analysis (1,000 replications per condition) using empirical 2PL parameters from PSN-IRT (Zhou et al., 2026) ($\bar{a} = 1.2$, $\bar{b} = -0.84$; 14,042 MMLU items) confirms that at $N \geq 80$ per cohort, ETS C achieves TPR of 0.68–0.80, FPR of 0.036–0.042, and AUC of 0.93–0.94 for items with $|\Delta_{\text{MH}}| \geq 1.4$ (details in Section A). Our temporal cohorts ($N_{\text{ref}} = 1,080$, $N_{\text{focal}} = 807$) far exceed this threshold.

4.2 Temporal Comparability: 31% Measurement Instability

We compare 2023 ($N = 1,080$) against 2024 ($N = 807$) models; 31.3% of items receive ETS C classification ($|\Delta_{\text{MH}}| \geq 1.5$; 95% bootstrap CI: $[29.4\%, 38.1\%]$, $B = 1,000$ model-level resamples). The rightward CI asymmetry reflects a heavy-tailed C% distribution across resamples, driven by subjects with C% exceeding 40%. Benjamini-Hochberg FDR correction at $q = 0.05$ removes only 3 items ($31.32\% \rightarrow 31.30\%$), because the dual ETS threshold on effect size and significance already provides implicit multiple-testing protection.

Three controls confirm this reflects genuine temporal change. A random split of the 2023 cohort yields only 0.18% ETS C items (sampling noise), and an ability-quintile comparison within the same temporal cohort yields 2.9% (ability differences alone). A within-family comparison (Llama-2 vs. Llama-3) produces 61.3% ETS C items, elevated by large ability gaps that confound the DIF signal (Holland and Thayer, 1988). The contrast between 2.9% (ability-controlled) and 61.3% (ability-confounded) brackets the temporal rate of 31.3%. Crucially, the temporal cohort ability gap is negligible: Cohen’s $d = -0.17$ (below the $|d| = 0.2$ small-effect threshold) with 76.4% distributional overlap. Under strict 1:1 nearest-neighbor matching ($d \approx -0.010$, $N = 638$ per group, 5 seeds), $\text{C}\% = 28.9\% \pm 0.1\%$ —a 7.7% relative reduction from 31.3%, confirming that at most ~ 2.4 pp of observed DIF is attributable to ability mismatch. A permutation test (200 label shuffles) yields null $\text{C}\% = 0.10\% \pm 0.12\%$ ($p = 0.005$; 0/200 exceed observed), confirming that 31.3% cannot be attributed to chance.

Across all 57 subjects, C% ranges from 20.7% to 48.4% (mean $\approx 30\%$; Table 1). DIF rates are domain-invariant (Kruskal-Wallis $p = 0.78$), but DIF *direction* exhibits domain dependence: STEM items favor 2024 models (55.4%), Humanities favor 2023 (64.4%). Item-level regression shows 92.4% of the DIF signal is item-specific (pseudo $R^2 = 0.0007$). A Simpson’s paradox confirms that DIF is associated with item properties rather than aggregate accuracy gains: domain accuracy positively predicts DIF at the aggregate level ($\text{OR} = 1.55$, $p = 0.001$) but reverses after controlling for item difficulty and discrimination ($\text{OR} = 0.73$, $p = 0.030$). The conclusion is robust across thresholds: $\text{C}\% = 48.9\%$ at $|\Delta_{\text{MH}}| \geq 1.0$, 18.3% at ≥ 2.0 , 10.2% at ≥ 2.5 ; ETS C’s dual threshold makes results less sensitive to the number of strata K than significance-only approaches ($K = 5$ per ETS practice (Donoghue and Allen, 1993)).

4.3 Ground Truth Mismatch

Direct comparison of DIF flags against the web-overlap labels of Li et al. (2024b) on 9,229 annotated items confirms that the two methods capture orthogonal signals. The enrichment ratio is 0.964 (Table 1): DIF-C rates are 30.7% among web-contaminated items and 31.8% among clean items. Association tests yield $\phi = 0.012$, $\chi^2 = 1.28$

Table 1: DIF landscape across cohort definitions. ETS C denotes the percentage of items with $|\Delta_{MH}| \geq 1.5$. FPR is measured against Li et al. web-overlap labels on the “clean” (non-overlapping) subset. Enrichment is the ratio of observed to expected overlap between DIF flags and web-overlap labels.

Cohort Definition	$N_{\text{ref}} / N_{\text{focal}}$	C%	FPR (clean)	Enrich.
Temporal (2023 vs. 2024)	1,080 / 807	31.3	0.318	0.964
Ability quintile (Q1–2 vs. Q4–5)	$\sim 2,600 / \sim 2,600$	2.9	0.027	1.19
Within-family (Llama-2 vs. 3)	161 / 308	61.3	–	–
Random split (sanity)	540 / 540	0.18	–	–
Within-subject (avg. 5 subj.)	<i>varies</i>	38.1	–	–

($p = 0.258$), and OR = 1.05 [0.96, 1.15]; the Jaccard index between flagged sets is 0.206. This near-independence holds across contamination subtypes (input-only enrichment = 0.986, input-and-label = 0.960). The null result does not indicate detection failure: semi-synthetic injection produces a clear dose-response ($\Delta\text{TPR} = +0.508$ at $p = 0.5$; Figure 2). The disconnect arises because web-overlap labels measure *global exposure* (whether an item exists in training corpora), while DIF measures *differential functioning* (whether an item’s relationship to ability changes across cohorts). The Li et al. labels are the wrong ground truth for a differential method. This near-independence extends to behavioral contamination detectors: the enrichment ratio of 0.964 and Jaccard overlap of 0.206 between DIF flags and web-overlap labels imply that any detector validated against these labels—including ConStat (Dekoninck et al., 2024)—is evaluated on a ground truth orthogonal to differential functioning.

4.4 Structural Analysis

Three structural controls fail to reduce temporal C% below 31% (Table 2): matched-ability quintiles, IRT θ -conditioning ($r(\theta, \text{total score}) = 0.9986$), and within-subject DIF (C% = 38.1%). Logistic regression DIF confirms 92.1% concordance with MH and reveals additional non-uniform DIF (24.8%), making 31% conservative. Graduated deduplication ($K=10$: 22.7%; $K=20$: 29.1%) confirms family structure does not inflate C%. Ability-matched analysis reduces Llama-2 vs. Llama-3 C% from 61.3% to 28.3% and temporal C% from 31.3% to 28.9%—consistent matched values (~ 28 –29%) despite vastly different unmatched rates, suggesting genuine item-level DIF accounts for ~ 28 –29% of items. Six alternative explanations (ability confounding, family inflation, θ -conditioning, within-subject effects, ranking impact, local dependence) were tested; none reduces C% below $\sim 28\%$

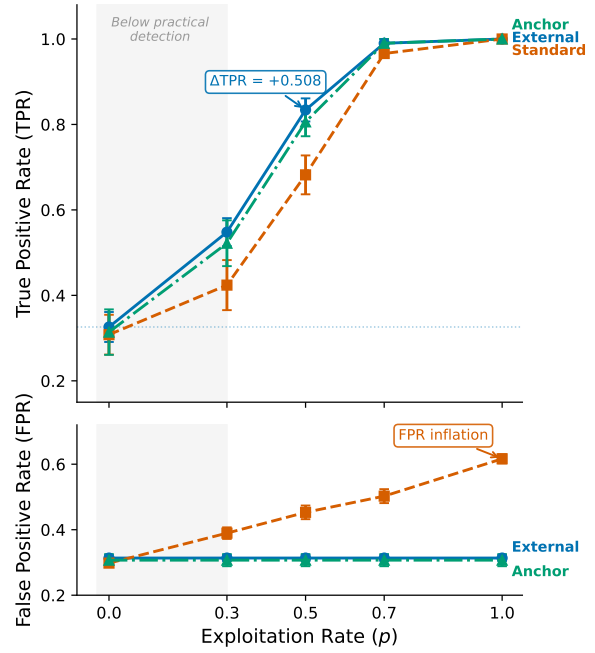


Figure 2: Semi-synthetic dose-response for DIF detection on MMLU. ΔTPR increases monotonically with exploitation rate p , confirming that MH-DIF is sensitive to differential item behavior when present. Baseline (no injection) produces $\Delta\text{TPR} \approx 0$.

Table 2: Structural approaches to reducing temporal DIF. None reduces temporal C% below 31%, indicating that the observed instability is not an artifact of ability matching.

Approach	Mechanism	Temp. C%	TPR ($p=0.5$)	Outcome
Baseline (ext.)	Cross-bench, total score	$0.314 \pm .009$	$0.834 \pm .027$	Reference
Matched-ability	Within-quintile temp. DIF	~ 0.31	–	Enrich. 0.94–1.04
θ -cond.	θ replaces total score	0.312	0.876	$r(\theta, \text{total}) = 0.9986$
Within-subject	Per-subject DIF	–	–	C% = 38.1% ($>25\%$)

(Section B).

4.5 Semi-Synthetic Framework Validation

Cross-subject matching achieves $\Delta\text{FPR} = 0.000$ across all five MMLU subjects (4/5 with $\Delta\text{TPR} > 0.3$ at $p = 0.5$), while within-subject matching inflates FPR by +0.186 on average. Multi-subject injection ($k \in \{1, 3, 5, 10, 20\}$ subjects, 5 random draws each) confirms robustness: $\Delta\text{FPR} < 0.007$ for $k \leq 5$ ($\leq 9\%$ of domains); at $k = 10$ (18%), ΔFPR rises to 0.026–0.069, marking the boundary of matching variable integrity (Section A and fig. 3).

4.6 Downstream Impact

Spearman correlation between full and stable-item rankings is $\rho \approx 0.997$: aggregate ordering is preserved despite 31% measurement instability. DIF-flagged items nonetheless concentrate discrimina-

Table 3: Temporal DIF across six METABENCH benchmarks. C% = ETS C items ($|\Delta_{MH}| \geq 1.5$) between 2023 and 2024 cohorts. All six show substantial DIF (45–174 $\times$ above random baseline), with knowledge/reasoning benchmarks exhibiting higher C% than procedural ones.

Benchmark	Items	C%	Rand. C%	Ratio
MMLU	12,508	31.3	0.18	174 $\times$
WinoGrande	1,267	30.3	0.32	95 $\times$
TruthfulQA	817	29.7	0.37	80 $\times$
ARC	1,172	26.9	0.60	45 $\times$
GSM8K	1,319	18.9	0.30	63 $\times$
HellaSwag	10,042	17.9	0.28	64 $\times$

tive signal ($z = -8\sigma$ to -49σ vs. random removal), though a circularity ablation confirms DIF-based removal is the *least* disruptive non-random strategy. Model families show distinctive DIF profiles: Llama-3 derivatives gain +53 median rank positions, Mistral loses 35 (Figure 4). Removing C-flagged items disproportionately disadvantages 2024 models (cohort gap = 96.2 positions, 40 $\times$ larger than random removal; $p < 0.001$), confirming that DIF items carry concentrated inter-generational signal.

4.7 Cross-Benchmark Validation

Temporal DIF extends across all six METABENCH benchmarks (Table 3). C% ranges from 17.9% (HellaSwag (Zellers et al., 2019)) to 31.3% (MMLU), each 45–174 $\times$ above random baselines. Knowledge/reasoning benchmarks (WinoGrande (Sakaguchi et al., 2020), TruthfulQA (Lin et al., 2022), ARC (Clark et al., 2018)) show C% = 26.9–31.3% versus 17.9–18.9% for procedural ones, consistent with knowledge-intensive items being more sensitive to training data composition shifts.

On GSM8K, semi-synthetic injection confirms detection power ($\Delta\text{TPR} = +0.514$ at $p = 0.5$; Section D). An exploratory GSM1K analysis ($N = 33$) shows a negative correlation ($r = -0.478$, $p = 0.005$) between contamination gap and DIF ranking advantage, suggesting DIF captures capability change distinct from memorization (Figure 5).

Even within a 6-month window (2023-H2 vs. 2024-H1), DIF remains substantial (MMLU C% = 34.35%, TruthfulQA C% = 31.33%).

5 Discussion

5.1 Implications for Benchmark Temporal Comparability

The 31% ETS C DIF rate extends across all six METABENCH benchmarks (C% = 17.9–31.3%, 45–174 $\times$ above baselines), with knowledge/reasoning benchmarks more affected than procedural ones. The Simpson’s paradox (Section 4.2) confirms DIF is associated with item properties rather than accuracy gains; ranking is preserved ($\rho \approx 0.997$) but measurement properties shift across model generations.

5.2 Ground Truth Meta-Critique

The near-independence between DIF flags and web-overlap labels (enrichment = 0.964, Jaccard = 0.206, $\phi = 0.012$) reflects that the two methods measure different constructs—global exposure versus differential functioning—making web-overlap labels an inappropriate ground truth for a differential method. The semi-synthetic dose-response ($\Delta\text{TPR} = +0.508$ at $p = 0.5$) confirms that MH-DIF detects differential behavior when present; the near-independence reflects a category error in the ground truth, not a detection failure.

5.3 Recommended DIF Audit Protocol

We propose that benchmark maintainers run temporal MH-DIF analysis whenever ≥ 500 new models accumulate, using cross-subject matching, with graduated action levels: (1) *Monitor* (C% < 15%); (2) *Flag* (15% $\leq$ C% < 25%): report stable-item rankings alongside full rankings; (3) *Act* (C% $\geq$ 25%): rotate or retire flagged items. MMLU’s C% exceeds the action threshold, aligning with ETS guidelines (Clauser and Mazor, 1998). DIF pre-screening identifies stable anchor items for equating (Habba et al., 2026); our open-source implementation runs in under 5 minutes on a single CPU core. Temporal C% could also serve as a dimension in composite quality indices (Zhu et al., 2026).

6 Conclusion

A semi-synthetic framework with cross-subject matching ($\Delta\text{FPR} = 0.000$) resolves a validation gap: web-overlap labels are near-independent of DIF flags (enrichment ≈ 0.96). Across six benchmarks, 17.9–31.3% of items exhibit temporal instability associated with item properties; LLM benchmarks need psychometric monitoring alongside score reporting.

Limitations

Methodological scope. MH-DIF detects only uniform DIF; logistic regression reveals an additional 24.8% non-uniform DIF, making 31% conservative. More fundamentally, DIF detects *that* items function differently but cannot determine *why*—the signal may reflect contamination, capability change, or training methodology shifts. A definitive causal decomposition remains out of reach without access to model training data. The semi-synthetic framework uses binary response flipping (incorrect $\rightarrow$ correct) to simulate differential contamination, representing a worst-case uniform exploitation pattern. Real-world contamination involves partial memorization, reasoning shortcuts, and format-specific advantages; detection power under such realistic patterns is untested. The moderate $\Delta\text{TPR} = +0.508$ at $p = 0.5$ reflects sensitivity to severe differential behavior; the framework’s primary methodological contribution—cross-subject matching achieving $\Delta\text{FPR} = 0.000$ —is a matching strategy property independent of injection realism.

Matching boundary and sample size. Cross-subject matching assumes contamination affects a minority of domains; when contamination spans >10 subjects ($>18\%$ of MMLU), ΔFPR rises to $0.026\text{--}0.069$. The GSM1K correlation analysis ($N = 33$) has limited statistical power and is exploratory; replication with larger model samples is warranted.

Generalizability. All six benchmarks use multiple-choice or binary scoring; generalizability to open-ended generation (e.g., AlpacaEval), coding (e.g., HumanEval), or partial-credit benchmarks is untested and would require polytomous DIF methods. Temporal cohorts are defined by calendar year, a coarse grouping; finer-grained definitions (e.g., by training data cutoff or alignment method) might reveal more targeted patterns.

References

- Simone Balloccu, Patrícia Schmidtova, Mateusz Lango, and Ondrej Dusek. 2024. [Leak, cheat, repeat: Data contamination and evaluation malpractices in closed-source LLMs](#). In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*.
- William Belzak. 2020. [Testing differential item func-](#)

[tioning in small samples](#). *Multivariate Behavioral Research*.

- Ryan Burnell, Wout Schellaert, John Burden, Tomer D. Ullman, Fernando Martinez-Plumed, Joshua B. Tenenbaum, Danaja Rutar, Lucy G. Cheke, Jascha Sohl-Dickstein, Melanie Mitchell, Douwe Kiela, Murray Shanahan, Ellen M. Voorhees, Anthony G. Cohn, Joel Z. Leibo, and Jose Hernandez-Orallo. 2023. [Rethink reporting of evaluation results in AI Science](#).
- Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. 2023. [Quantifying memorization across neural language models](#). In *International Conference on Learning Representations (ICLR)*.
- Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Kaijie Zhu, Hao Chen, Linyi Yang, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, Wei Ye, Yue Zhang, Yi Chang, Philip S. Yu, Qiang Yang, and Xing Xie. 2024. [A survey on evaluation of large language models](#). *ACM Transactions on Intelligent Systems and Technology*.
- Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios N. Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael I. Jordan, Joseph E. Gonzalez, and Ion Stoica. 2024. [Chatbot arena: An open platform for evaluating LLMs by human preference](#). In *Proceedings of the International Conference on Machine Learning (ICML)*.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.
- Brian E. Clauser and Kathleen M. Mazor. 1998. [Using statistical procedures to identify differentially functioning test items](#). *Educational Measurement: Issues and Practice*.
- Jasper Dekoninck, Mark Niklas Mueller, and Martin Vechev. 2024. [ConStat: Performance-based contamination detection in large language models](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- John R. Donoghue and Nancy L. Allen. 1993. An empirical examination of the Mantel-Haenszel statistic for the study of differential item functioning. *Applied Measurement in Education*, 6(2):163–179.
- Shahriar Golchin and Mihai Surdeanu. 2024. [Time travel in LLMs: Tracing data contamination in large language models](#). In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Eitam Habba, Ido Itzhak, Asaf Yehudai, Yotam Perlitz, Elron Bandel, Michal Shmueli-Scheuer, Leshem Choshen, and Gabriel Stanovsky. 2026. [Growing pains: Extensible and efficient LLM benchmarking via fixed parameter calibration](#). *Preprint*, arXiv:2604.12843.

- Matthew Hardy, Juan Gilbert, and Benjamin Domingue. 2026. [Efficient detection of bad benchmark items with novel scalability coefficients](#). *Preprint*, arXiv:2603.24999.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. [Measuring massive multitask language understanding](#). In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Paul W. Holland and Dorothy T. Thayer. 1988. [Differential item functioning and the Mantel-Haenszel procedure](#). In Howard Wainer and Henry I. Braun, editors, *Test Validity*. Lawrence Erlbaum Associates.
- Douwe Kiela, Max Bartolo, Yixin Nie, Divyansh Kaushik, Atticus Geiger, Zhengxuan Wu, Bertie Vidgen, Grusha Prasad, Amanpreet Singh, Pratik Ringshia, Zhiyi Ma, Tristan Thrush, Sebastian Riedel, Zeerak Talat, Pontus Stenetorp, and Robin Jia. 2021. [Dynabench: Rethinking benchmarking in NLP](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*.
- Seonghoon Kim and Michael Kolen. 2019. [Application of IRT fixed parameter calibration to multiple-group test data](#). *Applied Measurement in Education*.
- Alon Kipnis, Konstantinos Voudouris, Luca Maria Schulze Buschoff, and Eric Schulz. 2024. [metabench – a sparse benchmark of reasoning and knowledge in large language models](#). *Preprint*, arXiv:2407.12844.
- John Lalor, Hao Wu, and Hong Yu. 2019. [Learning latent parameters without human response patterns: Item response theory with artificial crowds](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*.
- Peiyang Li, Xiaobo Tang, Sai Chen, Ying Cheng, Ronald Metoyer, Tianyi Hua, and Nitesh V. Chawla. 2025. [ATLAS: Adaptive testing for LLM evaluation: A psychometric alternative to static benchmarks](#). *Preprint*, arXiv:2511.04689.
- Yucheng Li, Frank Guerin, and Chenghua Lin. 2024a. [LatestEval: Addressing data contamination in language model evaluation through dynamic and time-sensitive test construction](#). In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*.
- Yucheng Li, Yunhao Guo, Frank Guerin, and Chenghua Lin. 2024b. [An open-source data contamination report for large language models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*.
- Rongzhen Liang. 2026. [DVD: Detection via variance of generation distribution for variant contamination](#). *Preprint*, arXiv:2601.04895.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. [TruthfulQA: Measuring how models mimic human falsehoods](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 3214–3252.
- Frederic M. Lord. 1980. *Applications of Item Response Theory to Practical Testing Problems*. Lawrence Erlbaum Associates, Hillsdale, NJ.
- Zhimeng Luo, Lixin Wu, Adam Frisch, and Daqing He. 2025. [Measuring competency, not performance: Item-aware evaluation across medical benchmarks](#). *Preprint*, arXiv:2509.24186.
- Hotaka Maeda. 2025. [Finding words associated with DIF: Predicting DIF using LLMs and explainable AI](#). *arXiv preprint arXiv:2502.07017*.
- Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. 2024. [tinyBenchmarks: evaluating LLMs with fewer examples](#). In *Proceedings of the International Conference on Machine Learning (ICML)*.
- William Meredith. 1993. [Measurement invariance, factor analysis and factorial invariance](#). *Psychometrika*.
- Bengt Muthén. 1985. [A method for studying the homogeneity of test items with respect to other relevant variables](#). *Journal of Educational Statistics*.
- Ratna Nandakumar. 1993. Simultaneous DIF amplification and cancellation: Shealy-Stout’s test for DIF. *Journal of Educational Measurement*, 30(4):293–311.
- Qi Qian and Chengsong Huang. 2026. [Benchmark²: Systematic evaluation of LLM benchmarks](#). *Preprint*, arXiv:2601.03986.
- Nambury S. Raju. 1988. [The area between two item characteristic curves](#). *Psychometrika*.
- Pedro Rodriguez, Joe Barrow, Alexander Miserlis Hoyle, John P. Lalor, Robin Jia, and Jordan Boyd-Graber. 2021. Evaluation examples are not equally informative: How should that change NLP leaderboards? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4486–4503.
- Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. 2020. WinoGrande: An adversarial Winograd schema challenge at scale. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 8732–8740.
- Robin Shealy and William Stout. 1993. [A model-based standardization approach that separates true bias/DIF from group ability differences and detects test bias/DTF as well as item bias/DIF](#). *Psychometrika*.

905	Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo	2024. Min-K%++: Improved baseline for detect-	959
906	Huang, Daogao Liu, Terra Blevins, Danqi Chen, and	ing pre-training data from large language models.	960
907	Luke Zettlemoyer. 2024. Detecting pretraining data	<i>Preprint</i> , arXiv:2404.02936.	961
908	from large language models . In <i>Proceedings of the</i>		
909	<i>International Conference on Learning Representa-</i>		
910	<i>tions (ICLR)</i> .		
911	Hariharan Swaminathan and H. Jane Rogers. 1990. De-	Hao Zhou, Heng Huang, Zhenheng Zhao, Lin Han,	962
912	tecting differential item functioning using logistic	Haoyu Wang, Ke Chen, Min Yang, Weidong Bao,	963
913	regression procedures . <i>Journal of Educational Mea-</i>	Ji Dong, Bo Xu, Changshui Zhu, Huijun Cao, and	964
914	<i>surement</i> .	Tiejun Zhao. 2026. Lost in benchmarks? rethink-	965
915		ing large language model benchmarking with item	966
916	Yongding Tao, Tian Wang, Yihong Dong, Huanyu Liu,	response theory . In <i>Proceedings of the AAAI Confer-</i>	967
917	Kechi Zhang, Xiaolong Hu, and Ge Li. 2026. Detect-	<i>ence on Artificial Intelligence (AAAI)</i> .	968
918	ing data contamination from reinforcement learning		
919	post-training for large language models . In <i>Proceed-</i>	Longyuan Zhu, Hairan Hua, Linlin Miao, and Bing	969
920	<i>ings of the International Conference on Learning</i>	Zhao. 2026. Benchmark health index: A system-	970
921	<i>Representations (ICLR)</i> .	atic framework for benchmarking the benchmarks of	971
922		LLMs. <i>arXiv preprint arXiv:2602.11674</i> .	972
923	Jeanne A. Teresi and John A. Fleishman. 2007. Differen-	Bruno D. Zumbo. 1999. <i>A Handbook on the Theory and</i>	973
924	tial item functioning and health assessment . <i>Quality</i>	<i>Methods of Differential Item Functioning (DIF): Lo-</i>	974
925	<i>of Life Research</i> .	<i>gistic Regression Modeling as a Unitary Framework</i>	975
926		<i>for Binary and Likert-Type (Ordinal) Item Scores</i> .	976
927	Kushal Tirumala, Aram Markosyan, Luke Zettlemoyer,	Directorate of Human Resources Research and Eval-	977
928	and Armen Aghajanyan. 2022. Memorization with-	uation, Department of National Defense, Ottawa.	978
929	out overfitting: Analyzing the training dynamics of		
930	large language models . In <i>Advances in Neural Infor-</i>	A Method Details	979
931	<i>mation Processing Systems (NeurIPS)</i> .		
932		A.1 Diagnostics	980
933	Robert J. Vandenberg and Charles E. Lance. 2000. A	Three diagnostic procedures complement the pri-	981
934	review and synthesis of the measurement invariance	mary DIF analysis.	982
935	literature: Suggestions, practices, and recommen-		
936	dations for organizational research . <i>Organizational</i>	Local independence (Yen’s Q3). The MH proce-	983
937	<i>Research Methods</i> .	dure assumes that item responses are conditionally	984
938		independent given θ . Yen’s Q3 statistic measures	985
939	Clara Vania, Phu Mon Htut, William Huang, Dhara	residual correlations between item pairs after con-	986
940	Mungra, Richard Yuanzhe Pang, Jason Phang,	ditioning on the latent trait. We compute Q3 for all	987
941	Haokun Liu, Kyunghyun Cho, and Samuel R. Bow-	within-subject and cross-subject item pairs, with	988
942	man. 2021. Comparing test sets with item response	$ Q3 > 0.20$ as the threshold for local dependence.	989
943	theory . In <i>Proceedings of the 59th Annual Meeting of</i>	Elevated within-subject dependence would indicate	990
944	<i>the Association for Computational Linguistics (ACL)</i> .	that items within the same subject share variance	991
945		beyond what θ captures, which could inflate within-	992
946	Haonan Wang, Haoyang Li, Boeun Ko, and Hongyang	subject DIF rates.	993
947	Zhang. 2026. On the fragility of benchmark contami-		
948	nation detection in reasoning models . In <i>Proceedings</i>	Domain-specificity. We test whether temporal	994
949	<i>of the International Conference on Learning Repre-</i>	DIF concentrates in domains where models im-	995
950	<i>sentations (ICLR)</i> .	proved most. At the ecological level, we com-	996
951	Colin White. 2025. LiveBench: A challenging,	pute Spearman ρ between subject-level C% and	997
952	contamination-free LLM benchmark . In <i>Proceed-</i>	subject-level accuracy improvement from 2023 to	998
953	<i>ings of the International Conference on Learning</i>	2024. At the item level, logistic regression predicts	999
954	<i>Representations (ICLR)</i> .	item-level DIF C classification from domain-level	1000
955		accuracy change, with and without item property	1001
956	Carol M. Woods. 2009. Empirical selection of anchors	controls (difficulty, discrimination). This decom-	1002
957	for tests of differential item functioning . <i>Applied</i>	position quantifies how much of the observed in-	1003
958	<i>Psychological Measurement</i> .	stability is attributable to aggregate domain-level	1004
	Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali	gains versus item-specific behavioral change.	1005
	Farzaneh, and Yejin Choi. 2019. HellaSwag: Can a		
	machine really finish your sentence? In <i>Proceedings</i>	Instability decomposition. When domain accu-	1006
	<i>of the 57th Annual Meeting of the Association for</i>	racy improvement is included alongside item-level	1007
	<i>Computational Linguistics (ACL)</i> , pages 4791–4800.	controls, we compute the counterfactual C% that	1008
	Jingyang Zhang, Jingwei Sun, Eric Yeats, Yang Ouyang,		
	Martin Kuo, Jianyi Zhang, Hao Yang, and Hai Li.		

would remain after removing domain-level effects. The gap between observed and counterfactual C% measures the contribution of domain-level improvement to the total instability.

A.2 Statistical Considerations

Multiple testing. The ETS C classification imposes a dual threshold (effect size $|\Delta_{MH}| \geq 1.5$ and significance $p < 0.05$), which provides implicit control over false positives. As a robustness check, we apply Benjamini-Hochberg FDR correction at $q = 0.05$. The random-split sanity check provides an empirical estimate of the false positive rate under the null hypothesis.

Threshold sensitivity. We report results across a range of DIF thresholds ($|\Delta_{MH}| \in \{1.0, 1.5, 2.0, 2.5\}$) to verify that qualitative conclusions do not depend on the specific ETS C cutoff. The standard threshold of $|\Delta_{MH}| \geq 1.5$ corresponds to the ETS C definition used throughout the paper.

IRT model fit. For transparency, we report that the MMLU response matrix shows poor fit to a unidimensional 2PL model (expected for a 57-subject multidimensional benchmark). This does not affect MH-DIF validity, which relies only on total score stratification rather than parametric IRT assumptions.

Architecture heterogeneity. The METABENCH MMLU population consists of 99.98% decoder-only models (5,226 of 5,227). This near-complete architectural homogeneity eliminates architecture as a confound for DIF analysis. We verify this by excluding the single encoder-decoder model and confirming identical results.

B Structural Analysis Details

Three progressively refined approaches fail to reduce the temporal false positive rate below 31%, revealing a structural property of the data rather than a methodological limitation. Matched-ability quintile analysis restricts comparisons to models of similar total score and yields enrichment ratios of 0.94–1.04, with temporal FPR unchanged at approximately 0.31. IRT θ -conditioning with 20 ability strata achieves $r(\theta, \text{total score}) = 0.9986$ and $\text{FPR} = 0.312$, confirming that the external total score already captures nearly all ability variance relevant to DIF computation. Within-subject

DIF analysis on 5 representative subjects, intended to isolate subject-specific effects, finds mean C% = 38.1%, exceeding the 25% threshold that would indicate viable bifactor structure. Table 2 summarizes these results.

The temporal FPR reflects genuine item-level instability that persists regardless of how ability is measured or conditioned. Yen’s Q3 diagnostic reveals mild local dependence within subjects (11.3% of within-subject item pairs exceed $|Q3| > 0.20$, vs. 2.2% cross-subject; $5\times$ ratio, $p = 2.3 \times 10^{-9}$), which contributes to elevated within-subject DIF rates but is not the primary driver: the cross-subject matching protocol addresses this bias, and temporal C% persists under external matching (full Q3 distributions in Appendix Figure A2).

C Additional Figures

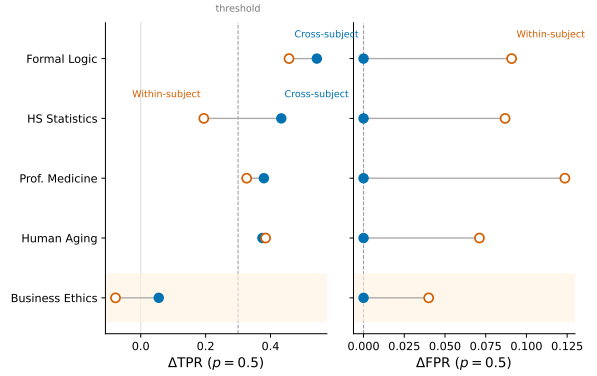


Figure 3: Cross-subject vs. within-subject matching for semi-synthetic DIF detection. Cross-subject matching eliminates FPR inflation ($\Delta\text{FPR} = 0.000$) while preserving detection power ($\Delta\text{TPR} > 0.3$ for 4/5 subjects at $p = 0.5$). Within-subject matching inflates FPR by +0.186 on average.

Table 4: Multi-subject injection robustness. k = number of simultaneously contaminated subjects (out of 57); values are mean $\pm$ SD across 5 random subject draws at exploitation rate $p = 0.5$.

k	ΔFPR	ΔTPR
1	$0.000 \pm .000$	$0.371 \pm .048$
3	$0.003 \pm .002$	$0.385 \pm .031$
5	$0.007 \pm .003$	$0.392 \pm .027$
10	$0.047 \pm .022$	$0.401 \pm .024$
20	$0.069 \pm .018$	$0.410 \pm .020$

D Cross-Benchmark Comparison

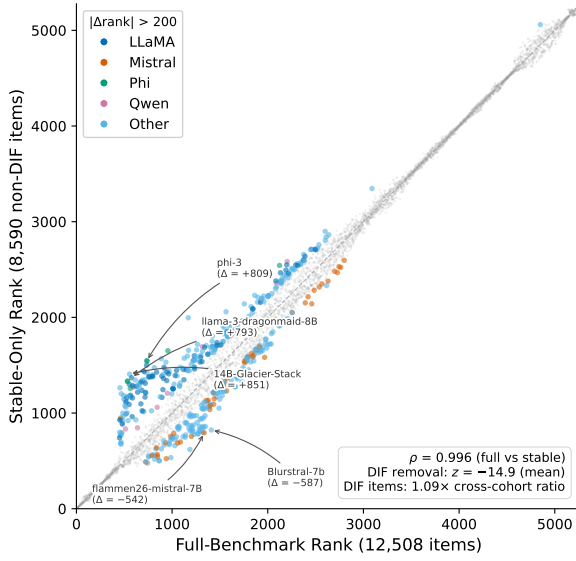


Figure 4: Downstream impact of DIF-based item filtering on model rankings. Despite $\rho \approx 0.997$ overall, individual model families show systematic rank shifts when evaluated on stable items only.

Table 5: Cross-benchmark comparison of DIF analysis on MMLU and GSM8K. Both benchmarks exhibit temporal measurement instability with validated semi-synthetic detection power. Standard and external matching use different exploitation rates (p) because each benchmark’s optimal operating point differs; external matching results are compared at $p = 0.5$ for both. Subscript $\pm$ values denote standard deviations across 5 random injection seeds.

Metric	MMLU	GSM8K
Items	12,508	1,319
Models	5,227	6,702
Temporal C%	31.3%	18.9%
Sanity C%	0.18%	0.3%
ΔTPR ($p=0.7$, standard)	—	$+0.818 \pm .006$
ΔTPR ($p=0.5$, external)	$+0.508 \pm .027$	$+0.514 \pm .028$

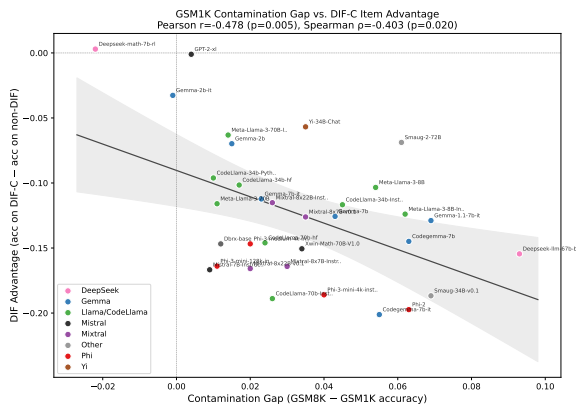


Figure 5: GSM1K contamination gap vs. DIF ranking advantage ($N = 33$). The negative correlation ($r = -0.478$, $p = 0.005$) indicates that models with larger contamination gaps benefit less from DIF-flagged items.